A CONVOLUTIONAL APPROACH TO MARKET PREDICTIONS

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Abstract.
Traditional approaches to Time Series Forecasting is regressive, utilizing data extrapolation techniques, creating a real-time numerical prediction using techniques such as recurrent and deep neural networks. Although regressive methods would be able to provide more insights into the market movement, they require a large amount of computational resources as there is a high level of complexity in training them to an acceptable accuracy. This paper presents a heuristic and 'visual' approach to market predictions, a technique using Residual Networks to analyse the graphs and create a probability distribution to model the stock movement with respect to the gradient. This method would hypothetically not require as much computational power to train as regressive tasks and be able to create a prediction accurate enough for market prediction, hence would be better for real-life applications such as high frequency trading. The S&P 500 Index Dataset has been used and predictions on this dataset has been made. It has been observed empirically that validation loss across all suitable models varies on a large extent, showing a high complexity for Residual Networks to learn the statistical dependencies within stock graphs, however the accuracy for the validation component is approximately 70% for the ResNet18 architecture.

Keywords.
Residual Networks, Convolutional Neural Networks, Financial Forecasting, Market Analysis, Technical Analysis

This technical report is to be submitted to the UCL Data Science Society (UCLDSS) and American Express for the 'American Express Challenge Virtual Hackathon', hosted by UCLDSS.

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INTRODUCTION

The challenge of predicting stochastic data has always been one of the most complicated tasks in history due to a large number of unprecedented changes to these data on a time series basis. An example of such stochastic data is the market - a set of data illustrating the volatility of the pricing for each type of 'currency'. Approaches such as Autoregressive Integrated Moving Average (ARIMA), Deep Neural Networks, Recurrent Neural Networks such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) have been introduced to model markets on a sequential basis, making the task a regressive task. However, these approaches consist of a flaw - they evaluate the market on a time-step to time-step basis, expecting the network to output a numerical representation of change in the market, represented as $f(\{x_1, x_2, \dots, x_n\}) = x_{n+1}$ causing the network to be complex and deep to be able to store the vast amount of statistical representation of the data. Extrapolation with the data might also be unreliable as what happened in the past might not constitute future actions.

The objective of this article is to present and show the feasibility of another approach to modelling stochastic data. In this regard, this paper aims to provide an alternative way to

market predictions, changing the subject of the task from a regressive task to a classification task, using image classifiers to predict the market. The most well-known deep-neural classification method would be Convolutional Neural Networks (ConvNet), which downscales images into feature maps and passes the feature maps onto a linear layer for processing. Amongst ConvNets, the residual architecture stands out most, for having multiple usages with images such as Super-Resolution such as SRResNet and SRGAN. For classification tasks, Residual Networks, or ResNets have proved most useful, and would be able to create a result within the shortest time as it is able to down-sample image into smaller feature maps whilst preserving statistical features.

The ResNet design mimics the human neural design in that intermediary layers would be skipped in layers such as the cortical layers in the human brain. This additive function between layers provides a similar function than the LSTM architecture - preventing vanishing gradients during backpropagation and information loss during forward propagation.

This paper introduces a heuristic method to predict the market, using Residual Networks to create a binary classification on whether if there would be a rise or fall in the next time step. By using this output from the network, it is expected that an investment strategy for a specific time-period would be returned depending on a hyperparameter named 'threshold'. The accuracy of the prediction would be calculated and an evaluation as to whether this method is suitable for high frequency trading would also be done.

RELATED WORK

There have been many attempts previously to predict the stock market. With machine learning techniques getting more advanced, it is imperative to rely on their power for further accuracy compared to simpler statistical modelling such as linear regression or ARIMA. [4] is one of the first projects to use neural networks for stock market prediction using artificial neural networks (ANN) to predict the Tokyo stock exchange index. Similarly, [6] had an overall prediction rate of 63% when they applied ANN to predict the buying and selling signals of the Tokyo stock exchange. In 1998, the attempts to predict the stock market were improved when [9] theorized two methods to solve the major problems that may occur during the training process: the use of momentum and starting the learning at random points. The efforts of [8] in 2000 have produced an accuracy of 81% when they trained a model genetically to predict the market direction of Singapore's stock exchange.

Previous statistical models made predictions based on linear models or traditional machine learning techniques have been replaced with ANNs. [1] found that although a specific price was not well predicted using ANNs, they nonetheless dramatically increased the accuracy of stock market trend predictions. ANNs prove to be the better method of market predictions due to their competence in understanding the complex relationship between the multiple parameters of the dataset and therefore producing a higher quality prediction to the stock trend. Linear models were unable to reach the level of complexity and generalisation that neural networks can achieve with their multiple inputs and dense complex connections between hidden layers. [1] made use of a standard backpropagation technique to train and test the data first assigning random weights to each node, eventually working towards optimal weights using gradient descent methods. They concluded that the simple ANN approach recorded a 98% training accuracy and the framework would have to be further improved upon for practical applications.

Nelson et al. [7] have approached the stock market prediction problem from a different angle. They have decided to use Long Short-Term Memory (LSTM), a form of recurrent neural networks known for its capabilities in Natural Language Processing (NLP). The plan was to use a large array of technical indicators to adequately assess the usage of LSTMs. Additionally, they wanted to test whether the short-term memory capability presents more accurate results than traditional feed forward networks. It was

concluded that using LSTM was able to provide them with high returns ratio per operation, meaning detecting high variation was done successfully, and that the system needs to improve before any more conclusions could be made.

Price based Machine Learning techniques of stock forecasting have greatly improved returns to stock traders who utilize technical analysis techniques over the last couple of decades. However, as these models have been used, improved and matured they have still proven to have a relatively large margin of error. This is because of the adverse effect that all the latest company news such as a merger, acquisition or a change in the board of directors *et cetera* can have on the speculation of the stock price and therefore on the demand and supply for the stock. [11] have found it advantageous to factor in company and general economic news along with existing statistical and deep learning methods to boost the accuracy and return even more reliable stock predictions. [11] has assumed that news only impacts stock the same day it is published and have grouped news into two main categories namely general market news and news that relates to the stock. Once pre-processed, a Support Vector Machine (SVM) [10] model was used for classifying the different documents. The document classified the sentiment as either 'good news' or 'bad news', this output was fed into another SVM to forecast the price. [11] conducted 5 sets of trials (price, direct news, indirect news, combined news and price and news) and compared their accuracies. It was interesting to note that the accuracy was highest (70.1%) for the technique that employed both the price and news rather than just the price (58.8%), direct news (62.5%) or combined news (64.7%) on their own. The net profit generated based on the predictions of each system nearly doubled with the combination of price and news rather than either of them separately when applying the SVM to a market simulation model. Although the combination of price and news returns higher benefits than either on their own, data fed into SVMs require feature selection as well as data pre-processing whereas the network in ResNets automatically pre-processes the data.

One of the newer machine learning algorithms is Convolutional Neural Network, a type of a feed-forward neural network mostly used for image classification. [2] implemented this model to predict the stock price movement of the Chinese stock market. They used a vector to represent their data along with an ADAM optimizer, a first order optimizer to optimize the hyperparameters in the kernel. Not only have they concluded that a CNN model is robust enough to make accurate predictions, the team has also explained how they intend to improve the system. CNNs and RNNs tend to have the competitive advantage for stock market predictions over regular neural networks simply because functions within their architecture would allow them to capture more statistical dependencies – RNNs being able to 'forget' and 'remember' key information using gates, and CNN would be able to extract feature maps within the dataset. However, using a large number of hidden layers in both architectures would not only require a large amount of resources, it also does not give back the same returns on accuracy which is why there is a need to introduce ResNets which uses larger kernels to select feature maps and adding the processed data to the pre-processed data between residual blocks, to downscale image faster and preventing information loss.

This architecture efficiently increases the accuracy in classification. For example, the ResNets have been proved useful in image classifying tasks such as CIFAR-10, having an 8.75% error with 0.27 million parameters, whilst the FitNet, with 2.5 million parameters, scoring a similar error, on 8.39%. [3]

It is unarguable that this architecture would efficiently classify images within a suitable timeframe better than any previously introduced methods such as Inception Nets and VGGNets.

METHODOLOGY

Stock Market movements are one of the most challenging prediction task. Technical analysis has stated that certain price movements may imply a rise or fall in the forthcoming time step. This observation skill can be taken on using

Residual Networks (ResNets). Hypothetically, the ResNet network should be able to select features within the graphical representation of the market.

The ResNet is expected to output a number between 0 and 1, created by the function 'sigmoid' in the final layer after a linear neuron in the final neural layer. This number denotes the probability of a rise or fall in gradient. The chosen architecture would be a ResNet. This study would also look into which Residual Network architecture would fit most for this task. The study is going to start with ResNet34 and exploit further into other models such as ResNet18 or ResNet50 depending on the training history of ResNet34.

Another feature deployed would be a deep fully connected linear layer rather than mapping the feature maps instantly to the final layer. This would be able to extract further information from the feature maps and therefore should hypothetically provide a higher accuracy.

The task for the Residual Network would be to create a probability distribution, therefore the objective of the Residual Network is to return 1 for an increase in price, and 0 for a decrease in price. This can be reached by computing a set of values which would satisfy:

$$\arg\min_{RN} -\log(RN(x)) \qquad (1)$$

The loss, therefore, is defined by the binary cross entropy loss - an indicator of how different two distinct probability distributions are. To reduce the loss, an efficient optimizer would be chosen to allow the gradient of the network to converge to a minima within a reasonable amount of time. Typical optimizers such as stochastic gradient descent have a disadvantage where the gradient is being scaled uniformly across all directions, whereas first order optimizers such as adaptive methods would not be able to converge to a minima in further timesteps. [5]

AMSBound, a method introduced in [5] would therefore be able to provide a better solution. As shown from one of the evaluation tasks done using AMSBound and ResNet34 on the CIFAR-10 dataset, AMSBound has performed better than SGD, ADAM, RMSProp, AdaBound, *et cetera*.

The AMSBound has a fundamental idea where it would change between the ADAM optimizer, to quickly reach a minima, and 'switch' to stochastic gradient descent to tune parameters on a fine scale.

To reach a good enough bias-variance trade-off, a total of 75 epochs for ResNet34 would be used. A graph illustrating the training history would also be presented in future chapters as a reference.

The dataset chosen would be the 'Standard & Poor 500' index, a reflection to the performance of the top 500 stock indices in the United States. S&P 500 is also categorized as one of the most representative market in the USA. Graphs are going to be generated with 30 data points, signifying that 30 minutes' worth of data would be used to generate a prediction for the next minute of whether the gradient will rise or fall. This can be tuned as one of the hyperparameters. By tuning the hyperparameter which indicates the number of minutes taken for a graph, it has been observed empirically that the graph would turn blurry on fine details. On an opposite case, if this hyperparameter is tuned to be a small number, there might not be enough datapoints in the graph for the network to generate an accurate enough

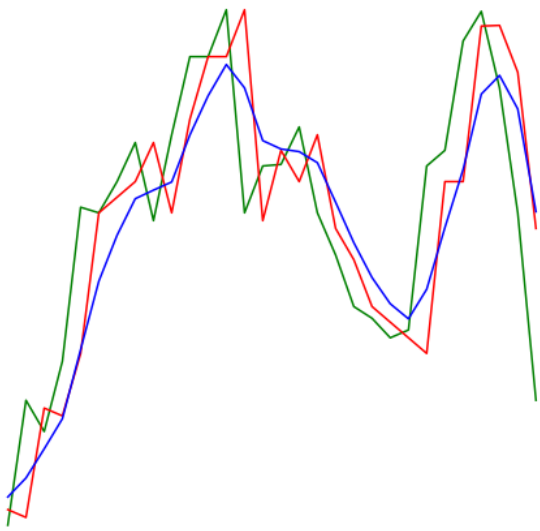

Fig.1: An illustration of the input data

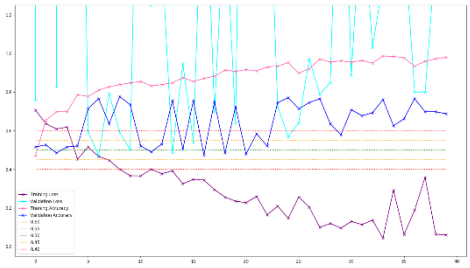

Fig. 2: ResNet18 training history

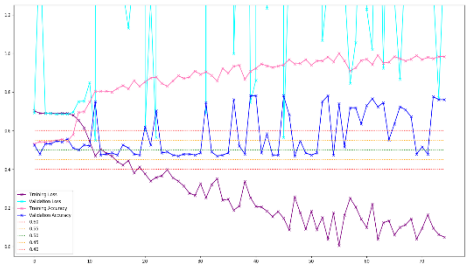

Fig. 3: ResNet34 training history

prediction. Fig. 1 illustrates a typical image generated to create a prediction.

The graph is going to be generated using the open price, the close price and the average price, with a rolling average of 3 minutes for the average price to reduce the amount of white noise within the data and create some difference between the graphs. The graphs are going to be generated into an 800 by 800 square image.

The data that is going to be used to generate the graph would not be scaled, but instead there will be no annotations to the graph. Making the image consist of nothing but the three graphs, as illustrated in Fig. 1. This would allow the ResNet to capture nothing but the stock movement.

RESULTS

This experiment is going to be ran under an NVIDIA GeForce RTX 2080 Ti. The CUDA acceleration that can be provided by the graphics card would be to accelerate the process of training than by an only CPU architecture by at least, four times. Using a CUDA architecture would also reduce the amount of forward propagation time, enabling a faster reaction to the market.

Upon investigation into the training history graph for ResNet34, it has been noticed that there are a lot of fluctuations with the validation loss across epochs, but not as much in the validation accuracy as shown in Fig. 3. This signifies that this task is a complex task to train and yet, there is an overall increase in the accuracy, reaching approximately 73% accuracy at the final epoch (75). This result is satisfactory for FinTech tasks.

To solve the problem of highly fluctuating validation loss, a hypothesis that the ResNet34 architecture overfits the data has been done. A ResNet18 architecture with lesser parameters than ResNet34, has therefore been used for this task to prevent overfitting. Since ResNet18 has lesser parameters, a smaller epoch count, such as 40 has been used to prevent overfitting.

The result of ResNet18 is much better than ResNet34, showing a promising 68.75% on the final epoch for the validation data, and an approximately linear increase in the training accuracy and a clear decrease in the training loss, as illustrated in Fig. 2. Although the validation loss is still fluctuating a lot, the volatility of this variable is not as exaggerated as the ResNet34 architecture. This architecture demonstrates a strong possibility to predict the market; this architecture used an approximate total of 0.15 seconds to create a graph and create a prediction, and therefore is suitable for high frequency trading.

DISCUSSION & FUTURE WORK

The authors have shown a possibility with market predictions using a classification approach with Residual Networks, achieving a 70% accuracy without hyperparameter fine tuning using a ResNet18. The authors have shown, in addition to the statistics, that ResNet34 overfits the two weeks of data in S&P 500 index, and therefore for all pre-evaluated architecture, the ResNet18 architecture would be the best architecture for this task.

Possibilities of stock prediction using convolutional neural networks have been shown on articles such as [2], reaching an accuracy of an approximate of 73% using Conv1D networks. This Conv2D approach has reached a worse accuracy, such as 68% for a ResNet18 architecture. However, considered that ResNet18 might have a chance of overfitting the model, as shown by the high variation in the validation loss, a residual model with lesser residual blocks than ResNet18 might be used to redo this task, to investigate whether the reduction of parameters in the task to optimize would reduce the volatility of the validation loss.

It is an intriguing discussion on which Residual designs should be used for these tasks; technical analysis is often used in high frequency trading purposes. This requires a low time complexity in forward propagation. Residual Networks and its variants such as ResNeXt can reduce the forward propagation time by reducing the size of the input data whilst retaining important statistical dependencies within them during forward propagation, and hence residual architectures have a low time complexity, however VGGNets are unable to do the same and therefore VGG has a higher time complexity.

It is also worth investigating into whether the background colour, the width of the graph lines and the colour of the lines chosen would adversely affect performance of the network. Colours such as black would hold a value of 0 for all the 3 channels and therefore might stand a better performance which is worth investigating into for further studies.

More architectures such as a 14-layered Residual Network might be designed and used which might perform better than the ResNet18 in terms of not overfitting the dataset, with a more stable validation loss.

This task can also be changed in perspective into an image recognition task rather than an image classification task, recognising features within the graphs using architectures such as YOLO and computing the probability of rise or fall based on the amount of features recognised using YOLO and the certainty that the features are correct.

A potential improvement to the algorithm can be such that instead of using a moving average line as one of the input, an autoencoder can be in place to clean the data up using neural techniques and therefore be processed by the convolutional neural network, reducing the complexity to predict the stock movement.

A large amount of state-of-the-art approaches such as AMSBound and ResNet18 has been applied for this specific report, and an appropriate hyperparameter optimization has not been done to the model yet. The authors believe that there are a lot of potential existing within the model for market predictions. It would also be interesting to see how the model behaves for Foreign Exchange Markets, or datasets with a higher level of stochastics.

CONCLUSION

The authors have presented a state-of-the-art approach to predict market performance using Residual Networks with an optimizer of AMSBound. The result of the investigation shows that the ResNet18 design fits well for two weeks of data but might still overfits the data whilst generating a 68.75% accuracy. An amount of future work, such as using a smaller residual network design, an investigation in using other image classifiers, or image recognisers can be done.

REFERENCES

[1] Abhishek, Kumar & Khairwa, A. & Pratap, T. & Prakash, Soam. (2012). A stock market prediction model using Artificial Neural Network. 2012 3rd International Conference on Computing, Communication and Networking Technologies, ICCCNT 2012. 1-5. 10.1109/ICCCNT.2012.6396089.

[2] Chen, Sheng & He, Hongxiang. (2018). Stock Prediction Using Convolutional Neural Network. IOP Conference Series: Materials Science and Engineering. 435. 012026. 10.1088/1757-899X/435/1/012026.

[3] Kaiming He, Xiangyu Zhang, Shaoqing Ren, & Jian Sun. (2015). Deep Residual Learning for Image Recognition.

[4] Kimoto T, Asakawa K Yoda M and Takeoka M (1990) Stock market prediction system with modular neural networks, in proceedings of the International Joint Conference on Neural Networks

[5] Liangchen Luo, Yuanhao Xiong, Yan Liu, & Xu Sun. (2019). Adaptive Gradient Methods with Dynamic Bound of Learning Rate.

[6] Mizuno H, Kosaka M, Yajima H and Komoda N (2001) Application of Neural Network to Technical Analysis of Stock Market Prediction

[7] Nelson, David & Pereira, Adriano & de Oliveira, Renato. (2017). Stock market's price movement prediction with LSTM neural networks. 1419-1426. 10.1109/IJCNN.2017.7966019.

[8] Phua, P.KE. Ming, D., Lin, W., Neural network With Genetic Algorithms for Stocks Prediction, Fifth Conference of the Association of Asian-Pacific Operations Research Societies, 5th -7th July (2000), Singapore.

[9] Sexton, Randall & Dorsey, Robert & Johnson, John. (1998). Toward global optimization of neural networks: A comparison of the genetic algorithm and backpropagation. Decision Support Systems. 22,. 171-185. 10.1016/S0167-9236(97)00040-7.

[10] V. N. Vapnik, The Nature of Statistical Learning Theory. New York: Springer-Verlag, 1995.

[11] Zhai, Yu & Hsu, Arthur & Halgamuge, Saman. (2007). Combining News and Technical Indicators in Daily Stock Price Trends Prediction. 1087-1096. 10.1007/978-3-540-72395-0_132